

Non-OCT Baseline for Oral Mucosa AI: Datasets and Papers Explained

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This document reviews the public datasets and papers that can serve as a **non-OCT baseline** for our oral soft-tissue CDSS work, plus the OCT papers that were shared alongside them. For each one it explains what the paper did, what data it provides, how to access it, what it is good for, its limitations, and what it means for our project.

1. What a non-OCT baseline is, and why it matters

The idea. Before claiming OCT helps, we should know what can already be done with the imaging clinicians use today: looking at the mouth (often photographed), and taking a biopsy for histology. A *baseline* is a system built on those existing methods. Later, the OCT system is judged against it: *does OCT add information a photograph cannot give?*

Where it fits in the project.

Work package part	Question	Data
WP2a (done)	Does our OCT measurement pipeline work?	Simulated OCT with known ground truth
WP2b (this document)	What does current non-invasive imaging achieve?	Public photos, multispectral images, histology
WP3	How does the real OCT system perform?	Partner OCT data
WP4	Demonstration CDSS	Can combine photo triage + OCT measurement

The one caution that shapes everything below. A fair head-to-head comparison needs the **same patients and the same lesions** imaged by both methods. The public photo datasets and our future OCT data come from different people, countries, devices and tasks. So:

- we **can** report “what existing non-invasive methods achieve on public data” as reference performance;
- we **cannot** claim “OCT beats photos” from unpaired data;
- for a true comparison, WP1 should ask OCT partners whether they also have **clinical photographs of the same scanned lesions**.

Also important: photos show the *surface* colour and texture of a lesion. OCT shows the *layers beneath*. They answer different questions, so the most realistic CDSS story is not “OCT replaces photos” but “photos flag a lesion, OCT measures what is under it.”

2. Overview

Of the four links shared, only MeMoSA is a non-OCT dataset. The other three are OCT papers — still very useful, and covered in Part D.

#	Paper / dataset	Modality	Size	Labels	Access	Main use for us
A1	MeMoSA (Chew et al., 2026)	Phone photos	30,039 images, 6,426 people, 5 countries	6 lesion types, 122 diagnoses, referral decision, risk habits, quality	Request via MeMoSA Workbench; CC BY / CC BY-NC	Main photo baseline; referral triage
A2	Song et al., 2021	Phone photos (cheek)	2,350 images	Normal / suspicious	On request	Uncertainty in oral image AI
A3	Fu et al., 2020	Clinical photos	44,409 images, 11 hospitals	Pathology-confirmed cancer vs normal	Not public	Reference performance at scale
A4	Piyarathne et al., 2024 (Sri Lanka)	Clinical photos	3,000 images, 714 patients	Healthy, benign, OPMD, cancer + boundaries	Zenodo, restricted, CC BY-NC-ND	Segmentation + classification baseline
A5	Ayman, 2024	Intraoral photos	9,201 images	Normal / low risk / high risk	Zenodo, non-commercial	Risk-level baseline
A6	SMART-OM, 2026	Phone photos	2,469 images, 331 subjects	4 classes + region annotations	Public (see paper)	Region-level baseline

#	Paper / dataset	Modality	Size	Labels	Access	Main use for us
A7	Kaggle oral cancer images	Photos (lip, tongue)	131 images	Cancer / non-cancer	Public	Not recommended (no methods/metadata)
A8	Sengupta et al., 2022	Review	—	—	—	Evidence that public oral data is scarce
B1	MODID multispectral dataset, 2024	Multispectral imaging	243 image cubes, 91 people	OSMF, leukoplakia, OSCC, healthy + segmentation	Public (Dryad)	Second optical baseline
C1	OCDC histology (Mendeley)	H&E slides	1,020 images	Tumour regions	Public	Histology reference
C2	Oral histopathology repository (Mendeley)	H&E slides	1,224 images	Normal epithelium vs OSCC	Public	Real epithelium appearance / thickness
D1	Dresden in vivo OCT, 2020	OCT (840 nm)	47 volunteers, 3,560 images	Healthy; thickness per site	Contact authors	Normative OCT values; partner
D2	Yuan et al., 2023 (<i>Oral Diseases</i>)	OCT	—	OSCC vs normal	Contact authors	OCT diagnosis comparator
D3	Yuan et al., 2024 (<i>BMC Oral Health</i>)	OCT, ex vivo	1,445 image sets, 68 patients	Prognosis, blood markers	On request	Related work only

3. Part A — Clinical photograph datasets

Clinical photographs are what a dentist or health worker can capture with a phone. They show colour, texture and shape of the mucosal surface. They are the most realistic “today’s practice” baseline.

A1. MeMoSA — the main photo baseline

Paper. Chew et al., *Scientific Data* 2026, from Cancer Research Malaysia and Universiti Malaya with partners in India, Sri Lanka, Indonesia and Nepal.

What it is. 30,039 oral mucosa images from 6,426 adults, collected 2015–2024 in Malaysia, India, Indonesia, Sri Lanka and Nepal. Images come from tertiary oral medicine/surgery clinics (12,264 images, 4,339 people) and community screening in Malaysian villages (17,775 images). All were taken on mobile phones with at least a 13-megapixel camera and flash, covering cheek, lip/sulcus, gum, floor of mouth, tongue and palate.

Labels. Each image has a clinical diagnosis (165 raw diagnoses unified into 122), grouped into **six lesion types**: oral cancer, oral potentially malignant disorder (OPMD), benign, developmental anomaly, normal anatomical variant, and no lesion. Each image also has a **referral decision**. Metadata includes age, sex, ethnicity, smoking, betel-quid chewing, alcohol, symptoms, medical history, and an **image-quality label** (good/poor with a reason: out of frame, artefact, out of focus, dark, overexposed).

How good are the labels? Only about 1% of images (323) have biopsy results. The rest were verified by dental experts; if a reviewer disagreed, a second expert reviewed and a consensus was reached. Teleconsultation images (57.6%) were standardised but not re-reviewed. So labels are mostly **expert clinical diagnoses, not histology** — important when interpreting any model trained on it.

Access. Register on the MeMoSA Workbench, get approved, submit a data-use request, and sign data-use agreements. Licences differ by source: Nepal, JSS India and Sri Lanka data are CC BY 4.0; Indonesia, Saveetha India and Malaysia data are CC BY-NC 4.0 (non-commercial).

Why it is valuable for us.

- The largest, most diverse public oral photo resource, with **referral decisions** — directly CDSS-relevant (“refer or not”).
- The **image-quality labels** mirror our own “quality gate” idea for OCT; a photo quality classifier is a natural parallel experiment.
- Risk-habit metadata (betel quid, smoking) allows risk-aware baselines.

Limitations. South and Southeast Asian populations only (generalisation to Irish/European patients is uncertain); mostly clinical rather than biopsy labels; phone photos vary in angle, light and focus; mostly non-commercial licences — check before any product use.

What it means for us. The best candidate for the WP2b baseline: train a photo model for (1) lesion vs no lesion, (2) referral vs no referral, and (3) image quality, and report calibrated uncertainty the same way as for OCT.

A2. Song et al., 2021 — uncertainty in oral photo AI

Used 2,350 phone-captured cheek images from three clinics in India, labelled normal or suspicious by oral oncology specialists, with **Bayesian deep learning** to estimate how reliable each prediction is. Data available on request. *Why it matters*: published in *Biomedical Optics Express* (our target journal) and about **knowing when an oral imaging model is unreliable** — the same theme as our uncertainty work. Cite it in the introduction and use its approach as a comparison for photo-baseline uncertainty.

A3. Fu et al., 2020 — large-scale photo detection

A deep-learning model detecting oral cavity squamous cell carcinoma from 44,409 clinical photos, pathologically validated, from 11 Chinese hospitals (*EClinicalMedicine*). Data not public. *Why it matters*: a reference point for what photo-based AI achieves at scale, and a reminder that the biggest datasets are private.

A4. Piyaathne et al., 2024 — Sri Lankan annotated dataset

3,000 images from 714 patients in Sri Lanka, labelled oral cancer, OPMD, benign or healthy, with **boundary annotations** of the oral cavity and each lesion, plus patient metadata (*Oral Oncology*). On Zenodo under restricted access, CC BY-NC-ND. *Why it matters*: lesion outlines allow a **segmentation** baseline, not just classification. *Limitation*: one hospital, one population; no-derivatives licence restricts redistribution.

A5. Ayman, 2024 — risk-level intraoral images

9,201 intraoral images labelled normal, low risk or high risk of malignant transformation, on Zenodo for non-commercial use. *Why it matters*: risk levels map neatly to a triage CDSS output. *Limitation*: check how labels were confirmed before relying on them.

A6. SMART-OM, 2026 — smartphone dataset with regions

2,469 smartphone images from 331 subjects in four classes (healthy, variation from normal, OPMD, cancer), with region-based and full annotations, fully anonymised (*Scientific Data*). *Why it matters*: a second, independent photo dataset for **external testing** of a photo baseline trained on MeMoSA.

A7. Kaggle “oral cancer (lips and tongue)”

The first public oral photo set (2020): 87 cancer and 44 non-cancer images, with no acquisition methods or metadata. *Not recommended* except as a quick sanity check; results on it are not credible evidence.

A8. Sengupta et al., 2022 — the scarcity paper

A letter in *Oral Oncology* documenting how few public oral cancer image datasets exist. *Why it matters*: cite it in the introduction to justify both the data-scarcity framing and our label-efficient approach.

4. Part B — Multispectral imaging

B1. MODID — multispectral oral disease images

What multispectral imaging is. A normal photo records three colours (red, green, blue). A multispectral camera records many narrow wavelength bands, forming a “data cube” (two spatial dimensions plus wavelength). Different tissue components (blood, collagen, keratin) absorb and reflect light differently at each wavelength, so the spectrum carries information about tissue composition that ordinary colour photos miss. It is still a *surface* technique — it does not give depth layers like OCT.

The dataset (*Scientific Data* 2024). 243 spectral data cubes from 91 participants (15 healthy, 76 with disease), captured without contact, covering buccal mucosa, labial mucosa, gingiva, tongue and palate. Conditions: **oral submucous fibrosis (OSMF)**, **leukoplakia**, and **oral squamous cell carcinoma**, with segmentation. Public on Dryad.

Why it is valuable for us.

- A second **optical, non-invasive** baseline, closer in spirit to OCT than phone photos.
- It includes **OSMF**, a collagen-stiffening disease where OCT (and polarization OCT) is expected to add value because the change is beneath the surface — a natural future comparison point.
- Same mouth sites as our OCT study.

Limitations. Small (91 people), single population, device-specific spectral bands; few healthy controls (15).

5. Part C — Histology datasets

What histology is. A biopsy is cut into thin slices, stained with haematoxylin and eosin (H&E), and viewed under a microscope. It shows the epithelium, the basement membrane and the lamina propria at cellular resolution. It is the **gold standard** for diagnosing dysplasia and cancer — and it shows exactly the layers OCT tries to see without cutting.

C1. OCDC — oral squamous cell carcinoma histology (Mendeley)

1,020 H&E images (640×640 pixels) of oral squamous cell carcinoma, with tumour regions manually labelled by a specialist and validated by a pathologist. Public.

C2. Histopathology repository of normal oral epithelium and OSCC (Mendeley)

1,224 H&E images at two magnifications; at 100×, 89 images of **normal oral epithelium** and 439 of oral squamous cell carcinoma. Public.

Why histology is useful for us (even though it is not a screening baseline).

- **Realism check for our simulator.** The normal-epithelium images show real rete-ridge shapes and boundary structure. If the pixel size is known (magnification alone is not enough — we need μm per pixel), we can measure real epithelial thickness and boundary undulation and compare with the simulator.
- **Understanding what an “unmeasurable” boundary looks like** in cancer: the loss of a clean epithelium–stroma boundary in OSCC is exactly what OCT later tries to detect.
- **A reminder about fixation.** Histology thickness is not the same as living-tissue thickness. The Dresden OCT study (Part D) notes that histology reports around 480 μm for buccal epithelium while in vivo OCT measured about 336 μm , attributing differences to tissue fixation and preparation. So histology thicknesses cannot be copied directly into OCT expectations.

Limitations. These datasets are for pathology tasks (tumour region detection), not measurement; slides come from different patients than any OCT data; pixel size must be confirmed before any thickness use.

6. Part D — The OCT papers that were shared

These are not non-OCT baselines, but they matter a lot for the OCT side of the project.

D1. Dresden in vivo OCT of healthy oral mucosa (Albrecht et al., *Diagnostics* 2020) — highly relevant

Who and where. Technische Universität Dresden, Germany (corresponding author: Julia Walther, julia.walther@tu-dresden.de).

What they did. Scanned **47 healthy adult volunteers** (25 men, 22 women, mean age ~29) with an **endoscopic OCT** system at 16 points in seven regions: labial, alveolar, buccal and sublingual mucosa, hard palate, soft palate/oropharynx, and palatine tonsils. In total **3,560 OCT images**. Smokers and heavy drinkers above tolerance levels were excluded.

Device. A spectrometer-based (spectral-domain) system at **840 nm**, 11.6 μm axial resolution in air, 17.4 μm lateral resolution, via a rigid Karl Storz endoscope; scanning was **without touching the mucosa** (contact-free), with a chin/head rest to reduce motion.

How they analysed images. A physician trained in OCT, with a histologist, scored each image on seven criteria: surface integrity, surface profile, epithelial homogeneity, **epithelial thickness** (mean of manual measurements, converted with refractive index 1.37), **basement membrane: intact / unsharp / not assessable**, vascularisation, and extra structures (salivary glands, lymphoid follicles). Histology slides from an anatomy collection (not the same people) were used for visual comparison.

Key findings (mean epithelial thickness). Sublingual ~120 μm (thinnest), alveolar ~142 μm , soft palate/oropharynx ~130 μm , tonsil ~125 μm , hard palate ~198 μm , labial ~243 μm , buccal ~336 μm (thickest). Men had thicker epithelium than women in the sublingual and palatal regions.

Findings that directly support our design.

- **“Cannot measure” is real and common.** In the buccal mucosa the basement membrane was **not assessable in 22.4%** of images because the thick epithelium exceeded the imaging depth. This is exactly why our system needs an explicit “cannot measure” output.
- **Keratinised sites are harder.** On the hard palate the boundary was **unsharp in 34.5%** of images, attributed to the keratinised surface reducing contrast. Our WP2 simulation independently found keratinised sites need their own uncertainty calibration — this paper gives real-world support.
- **Contact matters.** They note a contact-based study reported different buccal values, partly due to tissue compression; contact-free scanning avoids this but loses depth. This supports recording probe contact/pressure as a confounder.

- **Wavelength matters.** They recommend a **1300 nm** system for deeper structures — consistent with our physics reasoning.
- **Automation is the stated next step.** The authors explicitly point to automatic epithelial-thickness measurement as future work.

What it means for us.

1. **Update our simulator** to their measured site thicknesses (our current planning values, e.g. buccal 560 μm , are too thick compared with their 336 μm).
2. **Add Dresden to the partner list** — a European group with in vivo healthy multi-site OCT, exactly our first-paper scope.
3. **Cite in the introduction** for normative values and for the unmeasurable-boundary problem.

Limitations. Healthy young volunteers only; 840 nm system (shallower than 1300 nm); manual measurements by one analyst team; histology from different people.

D2. Yuan et al., 2023 — OCT diagnosis of oral squamous cell carcinoma (*Oral Diseases*)

From Beijing Stomatological Hospital (Capital Medical University). A **multi-level deep residual learning** model to automatically identify malignant tissue on OCT images, with performance compared across methods at image level and resected-patch level. The “resected patch” evaluation suggests tissue removed at surgery, as in the group’s later ex vivo work — to be confirmed in the full text. *Why it matters:* an OCT **diagnosis** comparator from the same group as D3; relevant to the later clinical-association stage, not to our measurement paper.

D3. Yuan et al., 2024 — OCT prognosis with blood markers (*BMC Oral Health*)

Already reviewed: **ex vivo** 850 nm OCT of 289 specimens from 68 patients (1,445 image sets), predicting recurrence using blood inflammation markers; data on request; code public. *Why it matters:* related work showing OCT + AI moving toward prognosis; also an example where image-level splits make evaluation hard to interpret — motivation for our patient-level design. Not usable as in vivo data.

7. How to use these for WP2b

Proposed baseline experiments

#	Experiment	Data	Output	Why
B-1	Photo triage: lesion vs no lesion, and refer vs not refer	MeMoSA (train), SMART-OM (external test)	Accuracy, sensitivity/specificity, calibration	What “today’s” non-invasive imaging achieves for referral
B-2	Photo quality gate	MeMoSA image-quality labels	Accept/reject with reason	Mirrors our OCT quality gate; reusable CDSS component
B-3	Uncertainty on photos	Same as B-1	Conformal sets for the photo classifier, coverage per country/site	Same trust framework as OCT; comparable with Song et al.
B-4	Lesion segmentation on photos	Piyarathne et al. (Sri Lanka)	Lesion outline	Where on the surface a lesion is — later the place to scan with OCT
B-5	Multispectral classification	MODID	OSMF / leukoplakia / OSCC / healthy	Second optical baseline, includes OSMF
B-6	Simulator realism check	Dresden OCT values; histology normal epithelium	Updated site thicknesses and boundary shapes	Makes the WP2 simulator closer to reality

What we can and cannot claim

Can claim:

- “On public photo data, a photo-based model achieves X for referral triage, with calibrated uncertainty.”
- “Our OCT pipeline and the photo baseline share the same quality-gate and uncertainty framework.”
- “Photos describe the surface; OCT measures layers beneath — the two are complementary in a CDSS.”

Cannot claim (without paired data):

- “OCT is more accurate than photographs.” Different patients, tasks and labels.
- That a photo model trained on South/Southeast Asian data will work in Ireland.

The paired-data ask for WP1

Add one question to every partner email: “*Do you also have clinical photographs of the lesions or sites you scanned with OCT?*” Even a small paired set turns the baseline into a genuine comparison.

How it feeds WP4 (demonstration CDSS)

A realistic demo flow: **photo → quality check → lesion flagged / referral suggestion → OCT scan of the flagged site → thickness with uncertainty or “cannot measure” → clinician review.** The photo baseline becomes the first step of the CDSS rather than just a comparison.

8. Access, licensing and action checklist

Dataset	How to get it	Licence notes
MeMoSA	Register at the MeMoSA Workbench, request access with intended use, sign data-use agreement	CC BY 4.0 (Nepal, JSS, Sri Lanka) / CC BY-NC 4.0 (Indonesia, Saveetha, Malaysia)
Piyarathne (Sri Lanka)	Zenodo request (DOI 10.5281/zenodo.10664056)	CC BY-NC-ND 4.0 — no derivatives, non-commercial
Ayman 2024	Zenodo (DOI 10.5281/zenodo.14571990)	Non-commercial
SMART-OM	See data availability in the paper	Check licence

Dataset	How to get it	Licence notes
MODID	Dryad (via the paper)	Check licence
OCDL, histology repository	Mendeley Data, direct download	Check licence
Song 2021, Fu 2020	Contact authors	Not public
Dresden OCT	Contact Julia Walther (julia.walther@tu-dresden.de)	Collaboration

Before using any dataset: read the data-use terms, record the licence in the project log, keep patient-level splits (MeMoSA provides patient IDs — use them), and do not redistribute restricted data.

Action checklist

1. Register on the MeMoSA Workbench and submit a research access request.
2. Download SMART-OM, MODID and the two histology sets.
3. Request the Sri Lankan and Ayman Zenodo datasets.
4. Update the WP2 simulator's site thicknesses using the Dresden values.
5. Add Dresden (TU Dresden) to the data-partner contacts document.
6. Add the paired-photo question to all partner emails.
7. Run B-1 to B-3 first (photo triage, quality gate, uncertainty) — they reuse the WP2 evaluation and conformal code directly.

9. References

- Chew S. et al. MeMoSA dataset: a multi-country collection of over 30,000 oral mucosa images with clinically labelled lesions. *Scientific Data* 13, 923 (2026). [Link](#) · Workbench: <https://workbench.memosa.my>
- Song B. et al. Bayesian deep learning for reliable oral cancer image classification. *Biomedical Optics Express* 12(10), 6422–6430 (2021).
- Fu Q. et al. A deep learning algorithm for detection of oral cavity squamous cell carcinoma from photographic images: a retrospective study. *EClinicalMedicine* 27 (2020).

- Piyrathne N.S. et al. A comprehensive dataset of annotated oral cavity images for diagnosis of oral cancer and oral potentially malignant disorders. *Oral Oncology* 156, 106946 (2024). [Link](#)
- Ayman N. An annotated clinical image dataset for deep learning-based classification of oral lesions. Zenodo (2024). <https://doi.org/10.5281/zenodo.14571990>
- SMART-OM: a smartphone-based comprehensive dataset of annotated oral cavity images. *Scientific Data* (2026). [Link](#)
- Sengupta N. et al. Scarcity of publicly available oral cancer image datasets for machine learning research. *Oral Oncology* 126, 105737 (2022).
- In-vivo non-contact multispectral oral disease image dataset with segmentation (MODID). *Scientific Data* (2024). [Link](#)
- OCDC: H&E-stained oral squamous cell carcinoma histological images dataset. Mendeley Data. [Link](#)
- Histopathological imaging database for oral cancer analysis. Mendeley Data. [Link](#)
- Albrecht M., Schnabel C., Mueller J., Golde J., Koch E., Walther J. In vivo endoscopic optical coherence tomography of the healthy human oral mucosa: qualitative and quantitative image analysis. *Diagnostics* 10(10), 827 (2020). [PMC7602587](#)
- Yuan W. et al. Noninvasive diagnosis of oral squamous cell carcinoma by multi-level deep residual learning on optical coherence tomography images. *Oral Diseases* 29(8) (2023). [Link](#)
- Yuan W. et al. Deep radiomics-based prognostic prediction of oral cancer using optical coherence tomography. *BMC Oral Health* 24, 1117 (2024). [Link](#)